

Physical Chemistry (I)

Lecture 07

Second Law of Thermodynamics



In the Last Class

- Enthalpy
 - Definition
 - Differential form of enthalpy
 - Heat capacities
 - Adiabatic changes
- Thermochemistry
 - Standard reaction enthalpy
 - Standard enthalpy of formation



Enthalpy H

$$H = U + pV$$

- Energy required to create a system (U) and to make a room for the system (pV)
- Meaning: $dH = \delta q_p$ or $\Delta H = q_p$

$$C_p = \left(\frac{\partial H}{\partial T} \right)_p$$

Last class

Differential Form of H

For $H = H(p, T)$,

$$dH = \left(\frac{\partial H}{\partial p} \right)_T dp + \left(\frac{\partial H}{\partial T} \right)_p dT$$

$$dH = -C_p \mu dp + C_p dT$$

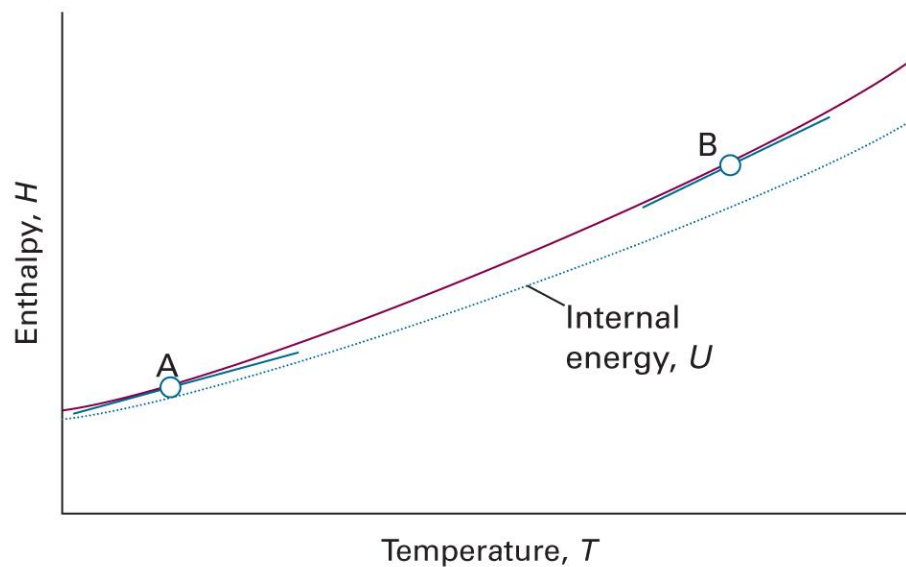
(Compare: $dU = \pi_T dV + C_V dT$)



Last class

C_V and C_p

$$C_V = \left(\frac{\partial U}{\partial T} \right)_V, \quad C_p = \left(\frac{\partial H}{\partial T} \right)_p$$



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 2B.3)

Last class

Adiabatic Changes

$$\delta q = 0$$

$$dU = \delta q + \delta w = \delta w$$

“Adiabatic expansion” $\rightarrow U$ drop $\rightarrow T$ drop

Last class

V - T and p - V Relationships

For $c = C_V/nR$,

$$V_i T_i^c = V_f T_f^c$$

$$VT^c = \text{constant}$$

For $\gamma = (c + 1)/c = 1 + 1/c$;

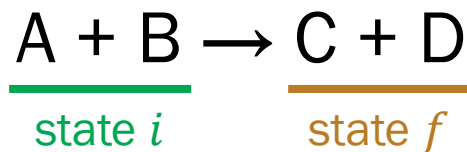
$$p_i V_i^\gamma = p_f V_f^\gamma$$

$$pV^\gamma = \text{constant}$$



Thermochemistry

“The study of the energy transferred as heat during the course of chemical reactions”



$$q_V = \Delta U$$

$$q_p = \Delta H$$

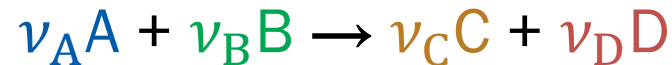
Standard Enthalpy, $H^\ominus$

Enthalpy H of a substance in its **standard state**

- **Standard state** of a substance: its pure form at 1 bar
 - Defined for a specific temperature
 - If temperature is not specified, use the conventional temperature of 298.15 K
- Standard enthalpy change $\Delta H^\ominus = H_f^\ominus - H_i^\ominus$

Last class

Standard Reaction Enthalpy



- Standard reaction enthalpy $\Delta_r H^\ominus$: standard enthalpy change $\Delta H^\ominus$ for the reaction per ...
 - ν_A moles of consumed A
 - ν_B moles of consumed B
 - ν_C moles of produced C
 - ν_D moles of produced D

Last class

Standard Enthalpy of Formation

- Standard enthalpy of formation $\Delta_f H^\ominus$ of a substance: $\Delta_r H^\ominus$ for the formation of the compound from its elements in their reference states
- Reference state of an element: its most stable state at the specified temperature and 1 bar

4. Entropy

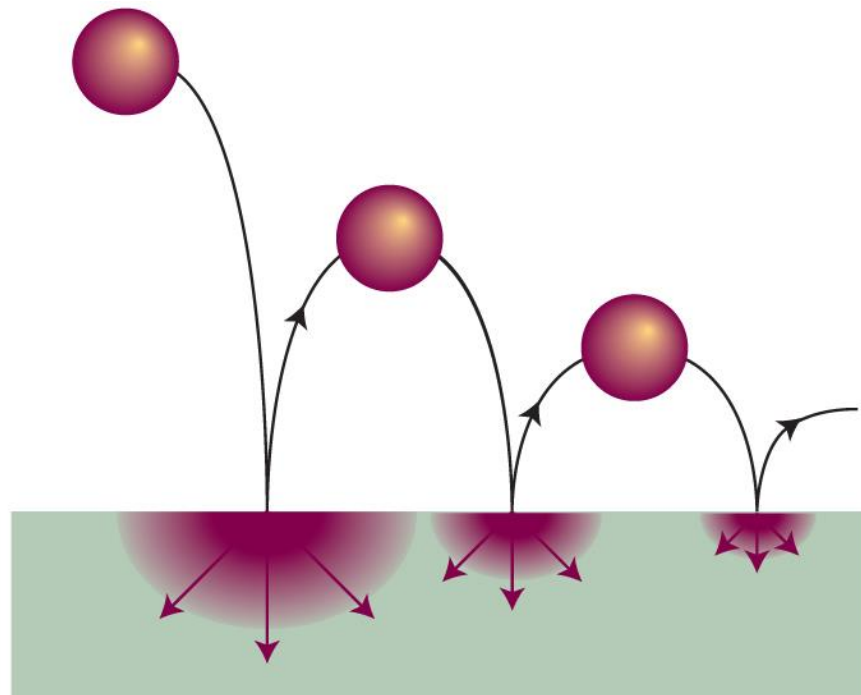


Chapter Overview

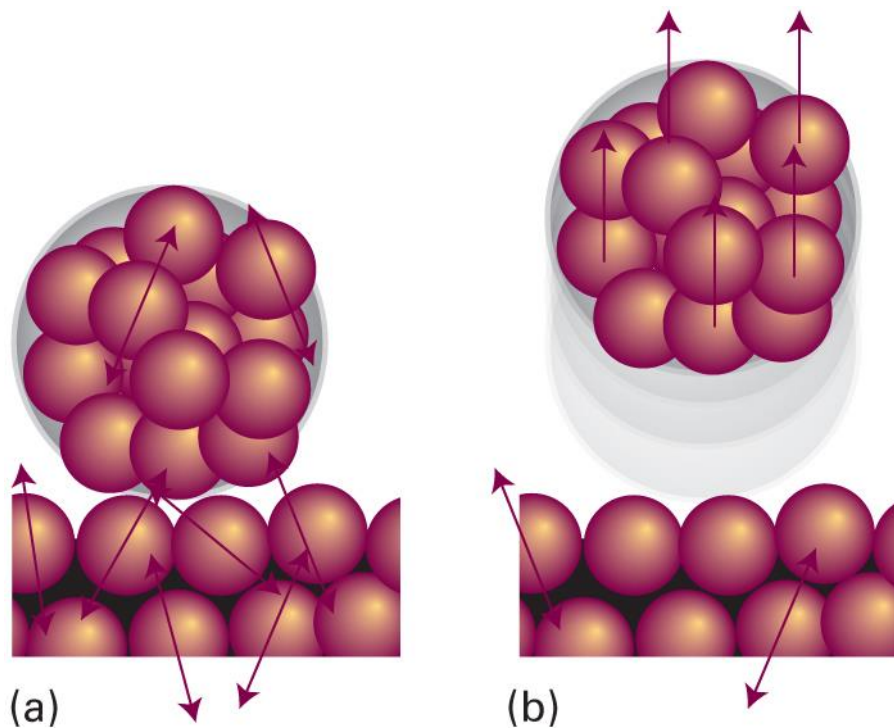
- Second law of thermodynamics
 - Kelvin's statement and Clausius's statement
 - Heat engine and Carnot cycle
- Entropy
 - Definition and properties
 - Third law of thermodynamics
 - Entropy and thermochemistry
- Physical meaning of entropy
 - Thermodynamic meaning
 - Statistical meaning: Boltzmann's formula



“Arrow of Time”

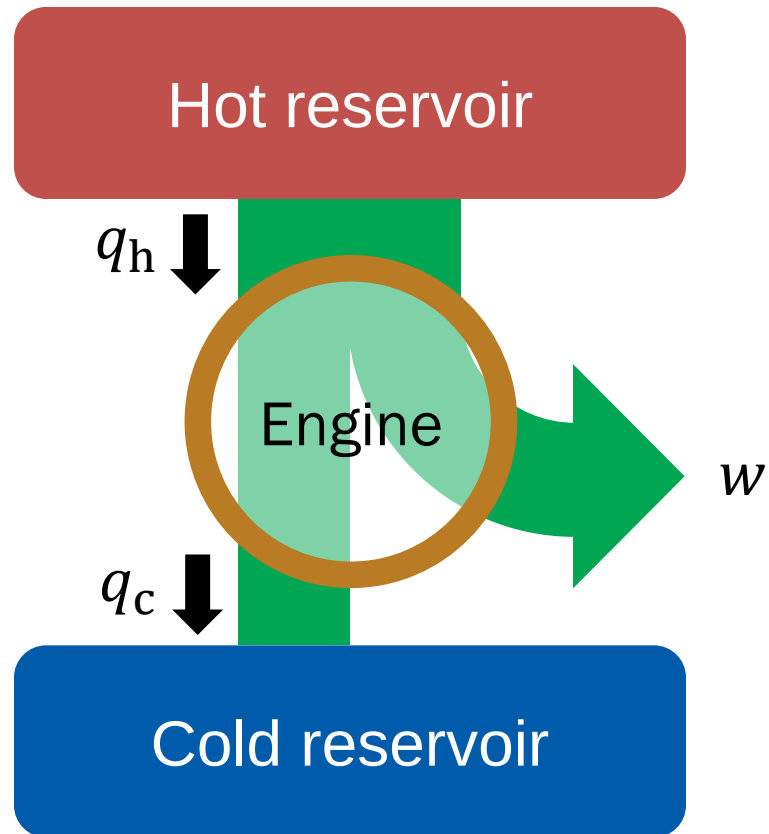


Why?



Heat and work are not equal

Heat Engine

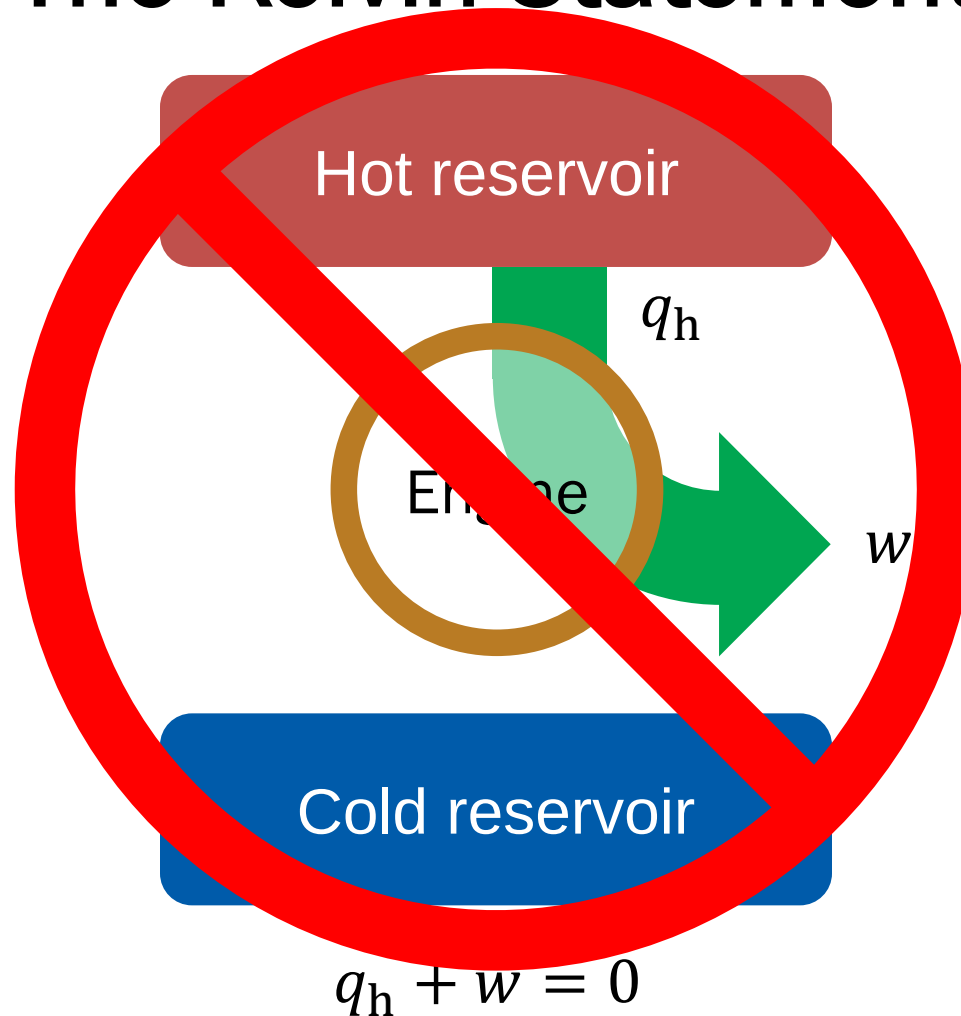


$$q_h + q_c + w = 0$$

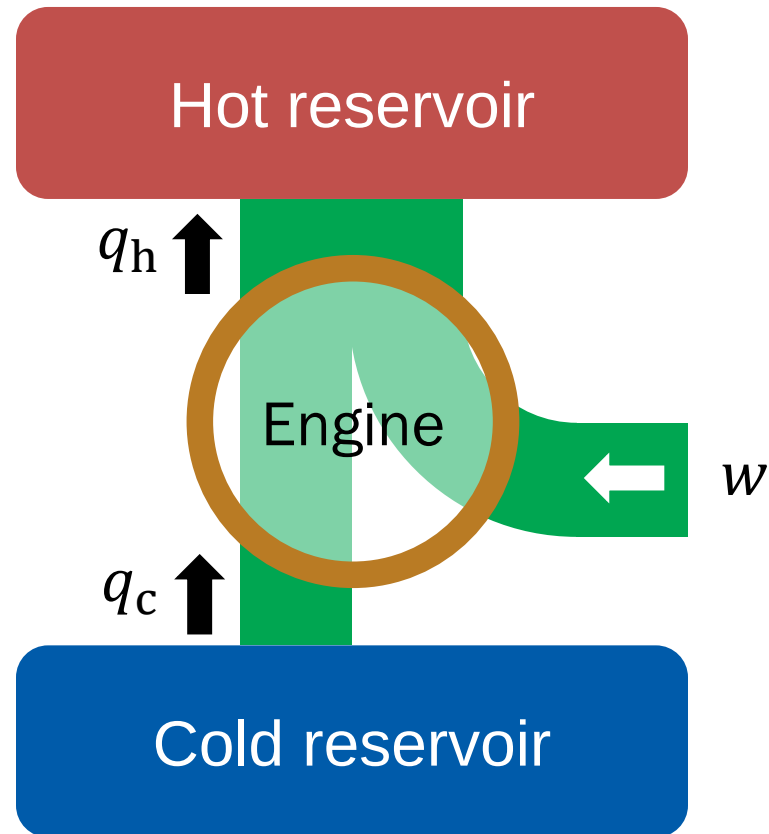
Second Law (Kelvin)

“No process is possible in which the sole result is the absorption of heat from a reservoir and its complete conversion into work.”

The Kelvin Statement



“Refrigerator”

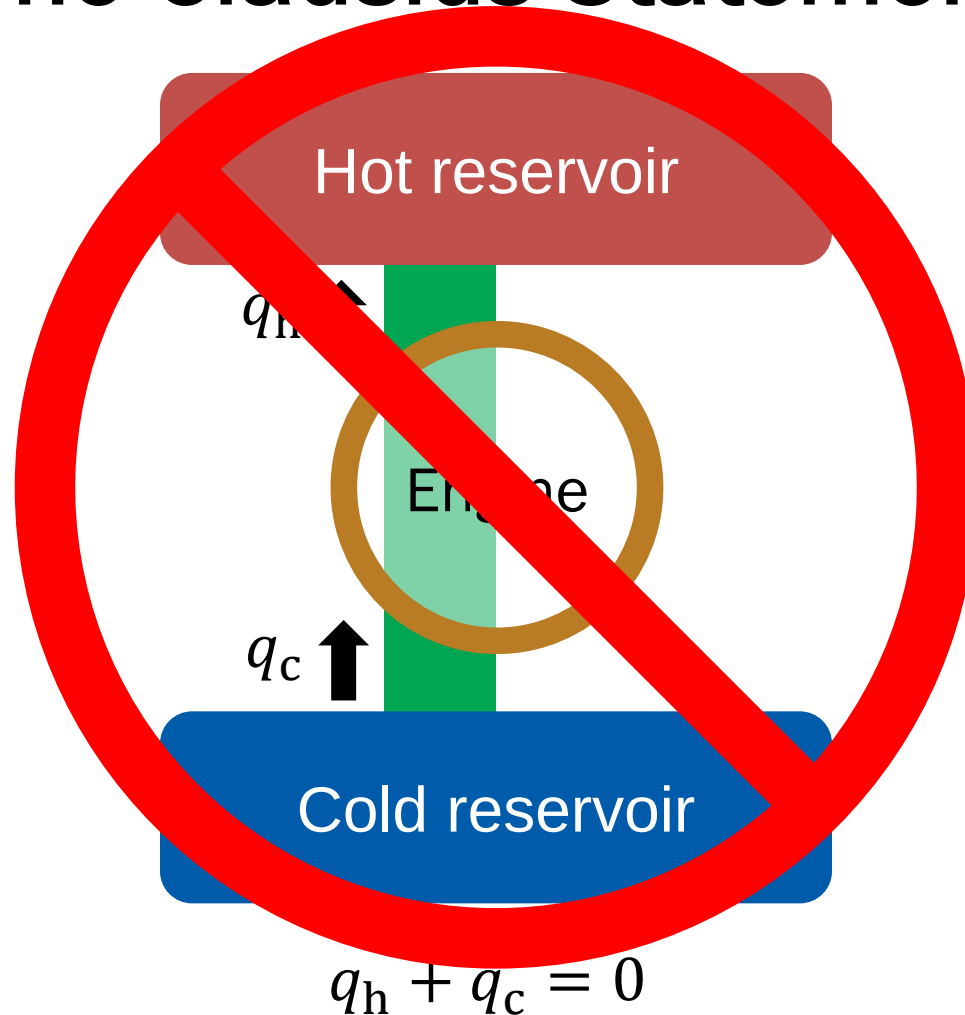


$$q_h + q_c + w = 0$$

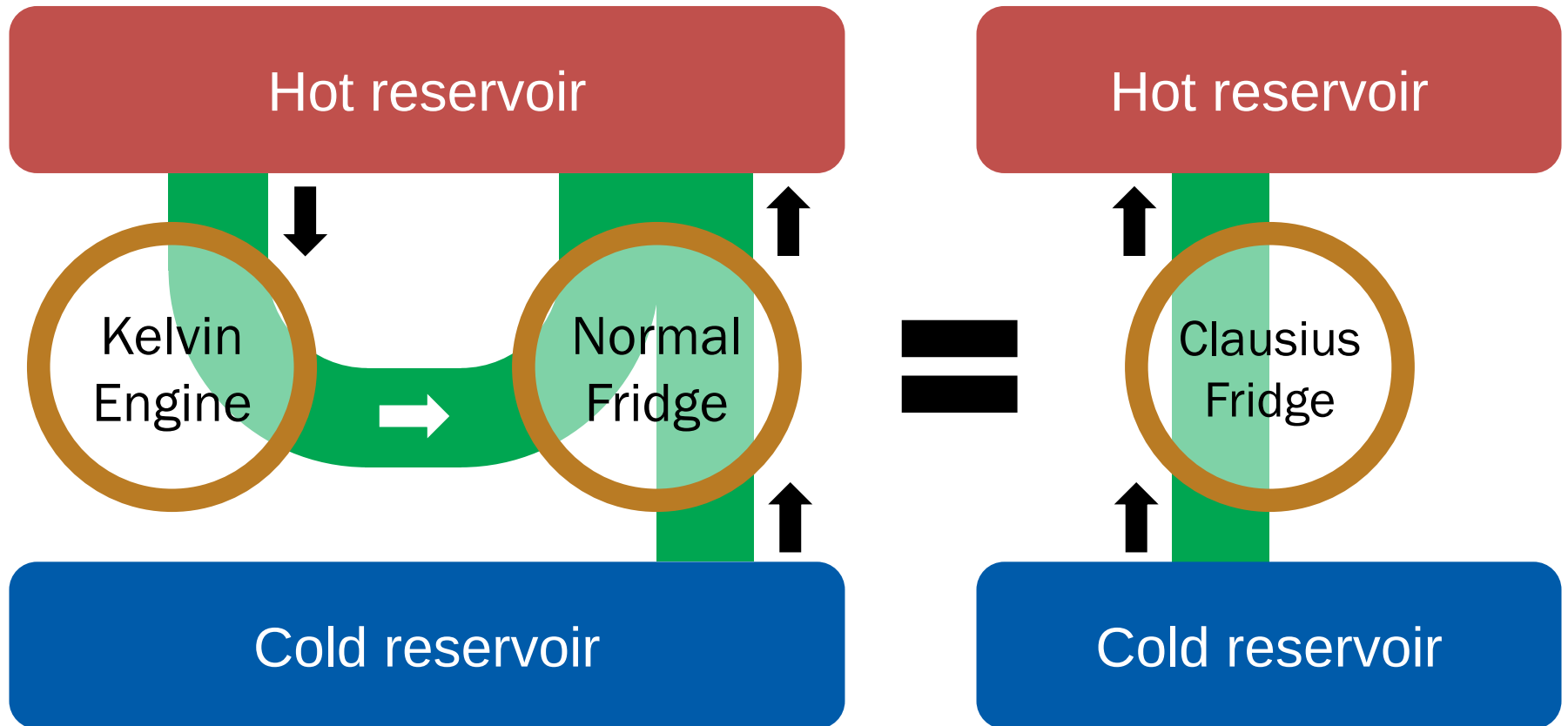
Second Law (Clausius)

“Heat does not flow spontaneously from a cool body to a hotter body.”

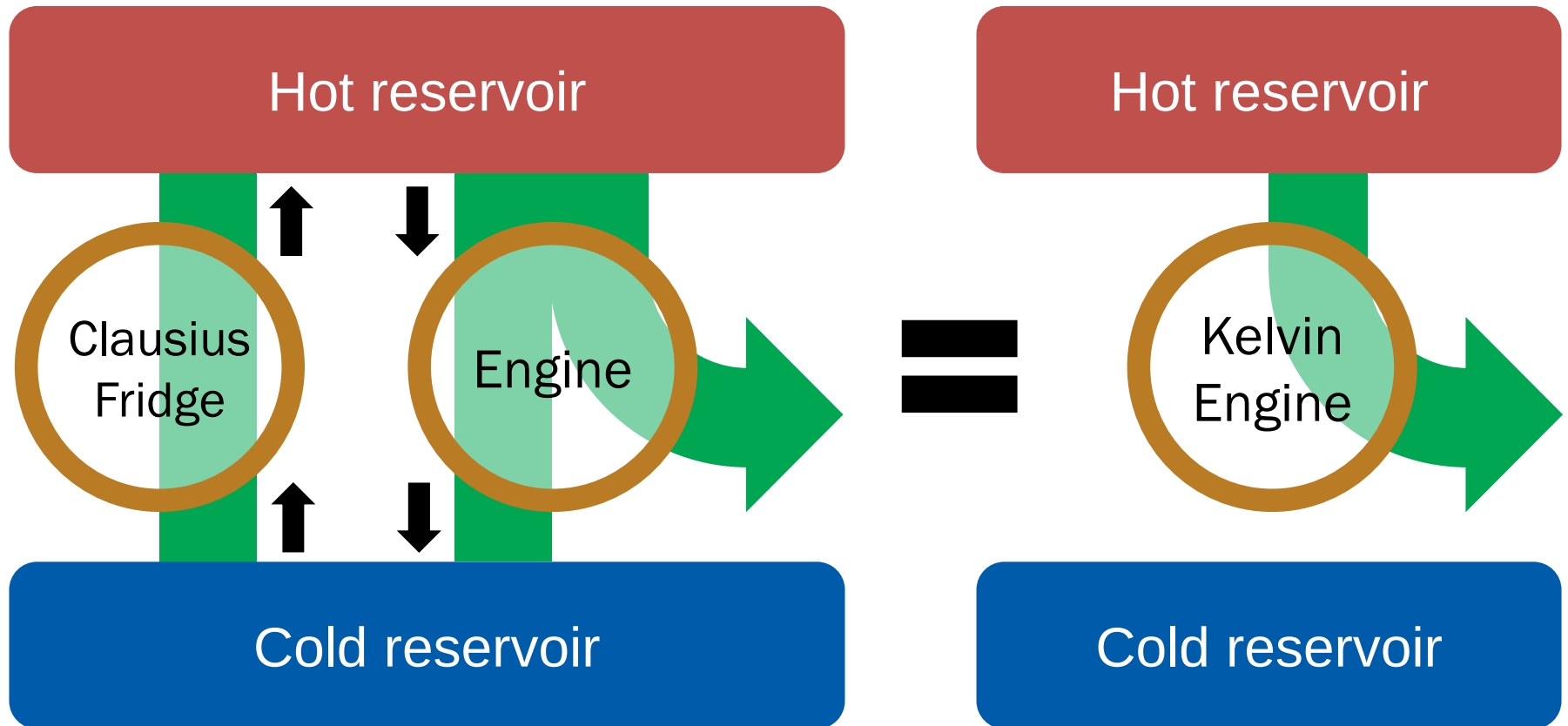
The Clausius Statement



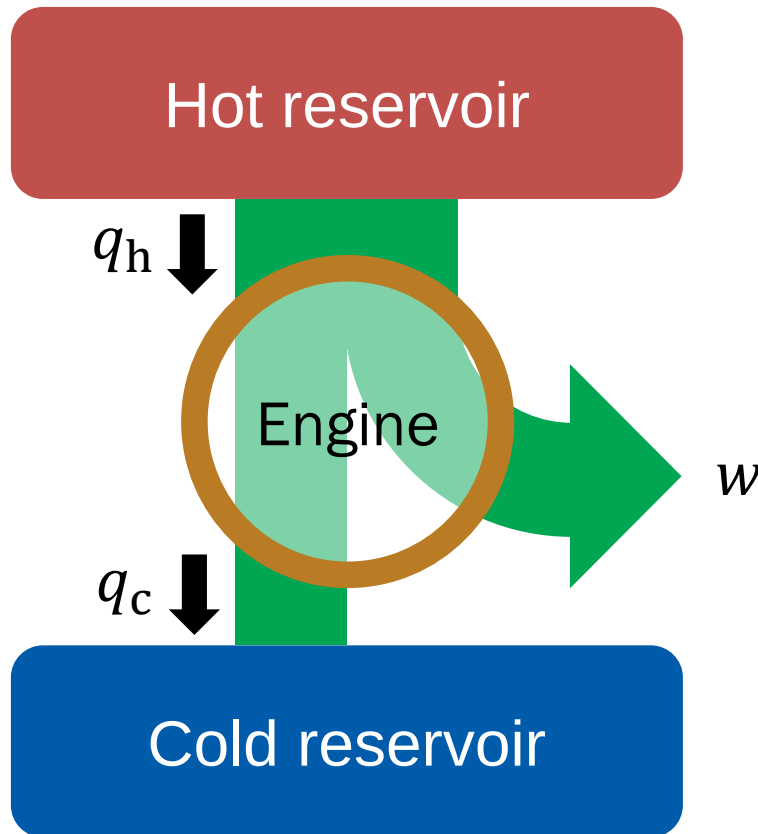
Kelvin $\rightarrow$ Clausius



Clausius $\rightarrow$ Kelvin



Efficiency of the Engine



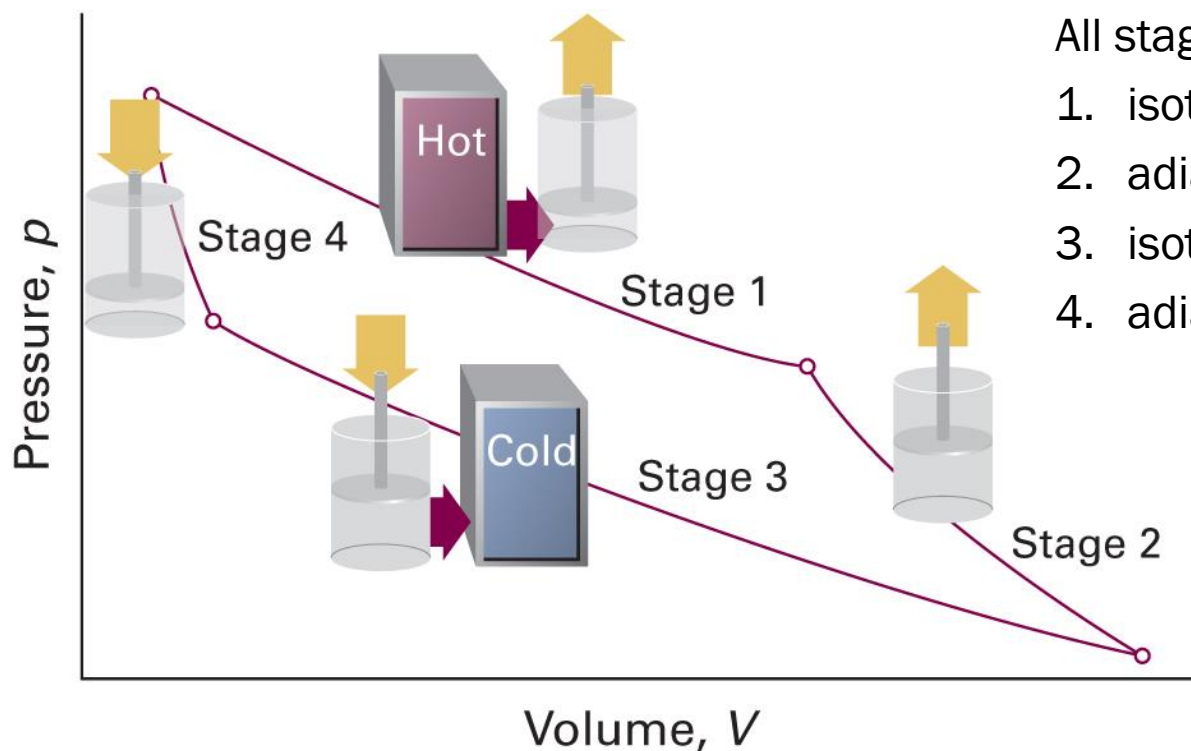
$$\eta = \frac{|w|}{|q_h|}$$

Hence,

$$\eta = \frac{|q_h| - |q_c|}{|q_h|} = 1 - \frac{|q_c|}{|q_h|}$$

$$q_h + q_c + w = 0$$

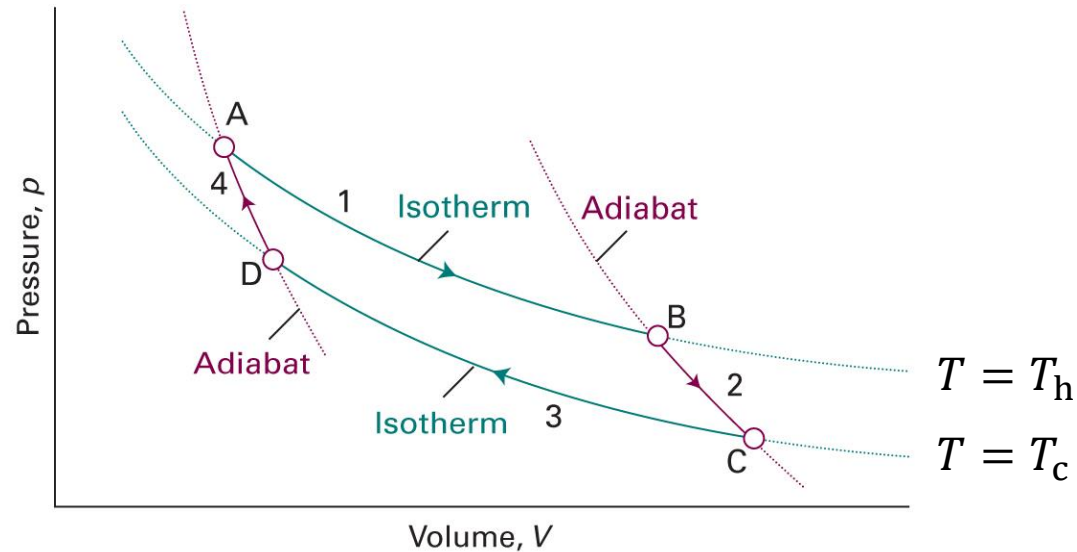
Carnot Cycle



- All stages are reversible;
1. isothermal expansion
 2. adiabatic expansion
 3. isothermal compression
 4. adiabatic compression

Not in Atkins

Perfect Gas in Carnot Engine



1. Isothermal expansion (at T_h)

$$w_1 = -nRT_h \ln(V_B/V_A)$$

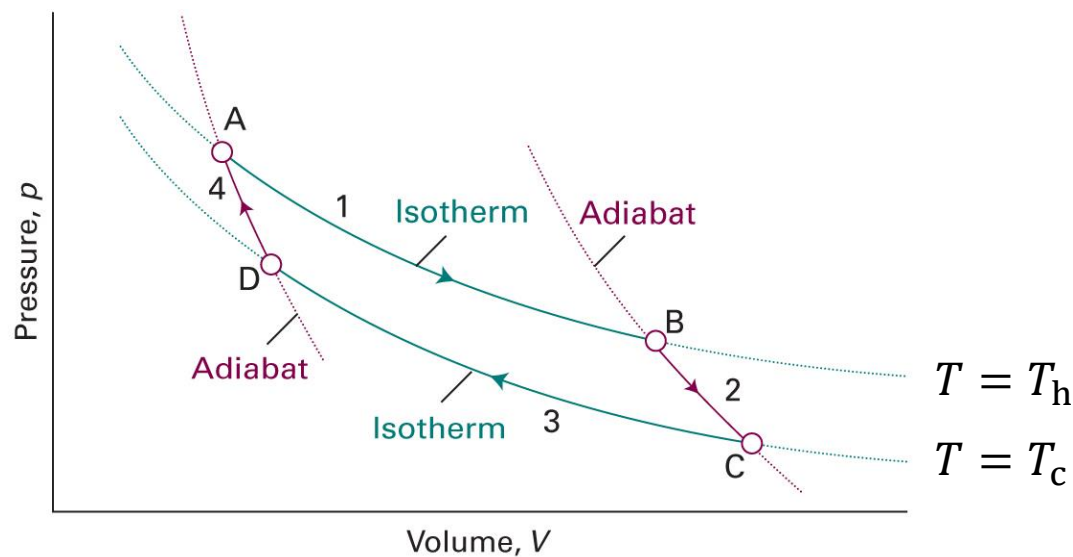
$$\text{Since } \Delta U = 0, q_1 = nRT_h \ln(V_B/V_A)$$



(Image: Atkins et al., *Atkins' Physical Chemistry*, 12th ed., Figure 3A.9)

Not in Atkins

Perfect Gas in Carnot Engine



2. Adiabatic expansion

$$q_2 = 0$$

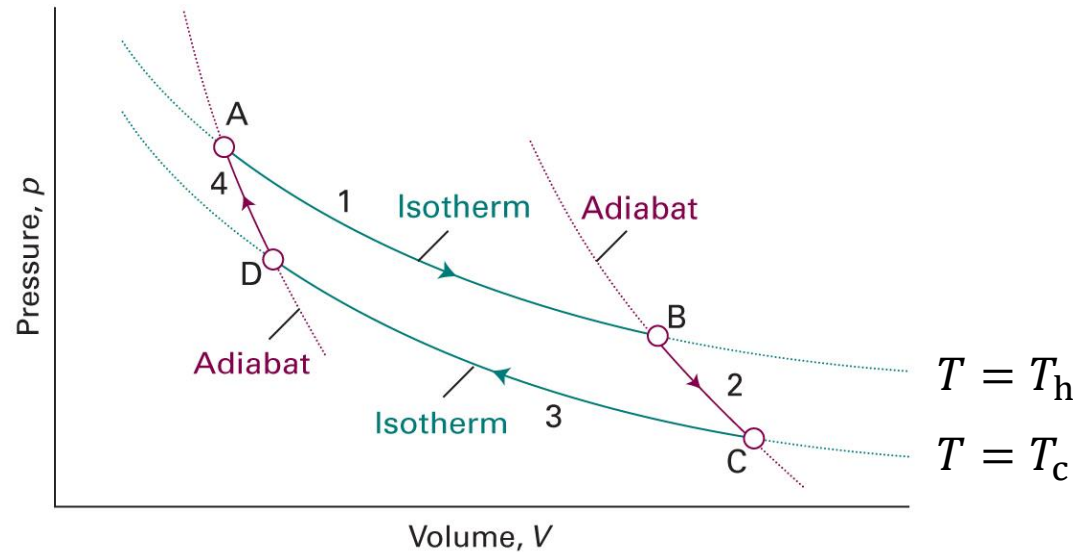
$$w_2 = C_V(T_c - T_h)$$



(Image: Atkins et al., *Atkins' Physical Chemistry*, 12th ed., Figure 3A.9)

Not in Atkins

Perfect Gas in Carnot Engine



3. Isothermal compression (at T_c)

$$w_3 = -nRT_c \ln(V_D/V_C)$$

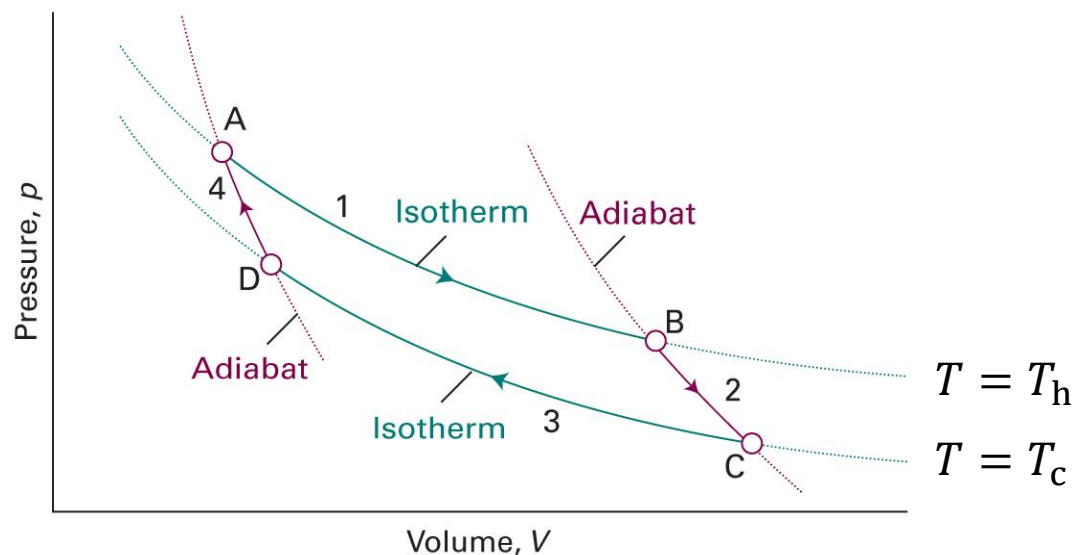
$$\text{Since } \Delta U = 0, q_3 = nRT_c \ln(V_D/V_C)$$



(Image: Atkins et al., *Atkins' Physical Chemistry*, 12th ed., Figure 3A.9)

Not in Atkins

Perfect Gas in Carnot Engine



4. Adiabatic compression

$$q_4 = 0$$

$$w_4 = C_V(T_h - T_c)$$



(Image: Atkins et al., *Atkins' Physical Chemistry*, 12th ed., Figure 3A.9)

After One Cycle ...

- Internal energy change $\Delta U = 0$ (state function)
- Enthalpy change $\Delta H = 0$ (state function)

- Heat

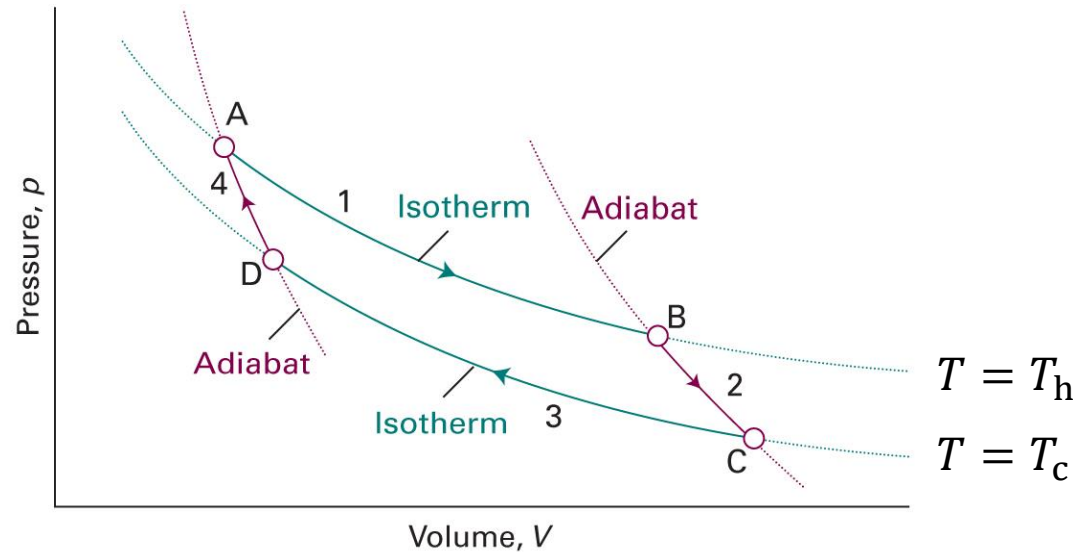
$$q = q_1 + q_2 + q_3 + q_4 = nRT_h \ln(V_B/V_A) + nRT_c \ln(V_D/V_C)$$

- Work

$$w = w_1 + w_2 + w_3 + w_4$$

$$w = -nRT_h \ln(V_B/V_A) - nRT_c \ln(V_D/V_C) = -q$$

Use of Adiabats



2. Adiabatic expansion: $V_B T_h^c = V_C T_c^c$ ($c = C_V/nR$)

4. Adiabatic compression: $V_D T_c^c = V_A T_h^c$

Heat and Work

Since $V_B T_h^c = V_C T_c^c$ and $V_D T_c^c = V_A T_h^c$,

$$\frac{V_B}{V_C} = \left(\frac{T_c}{T_h} \right)^c = \frac{V_A}{V_D}$$

Hence,

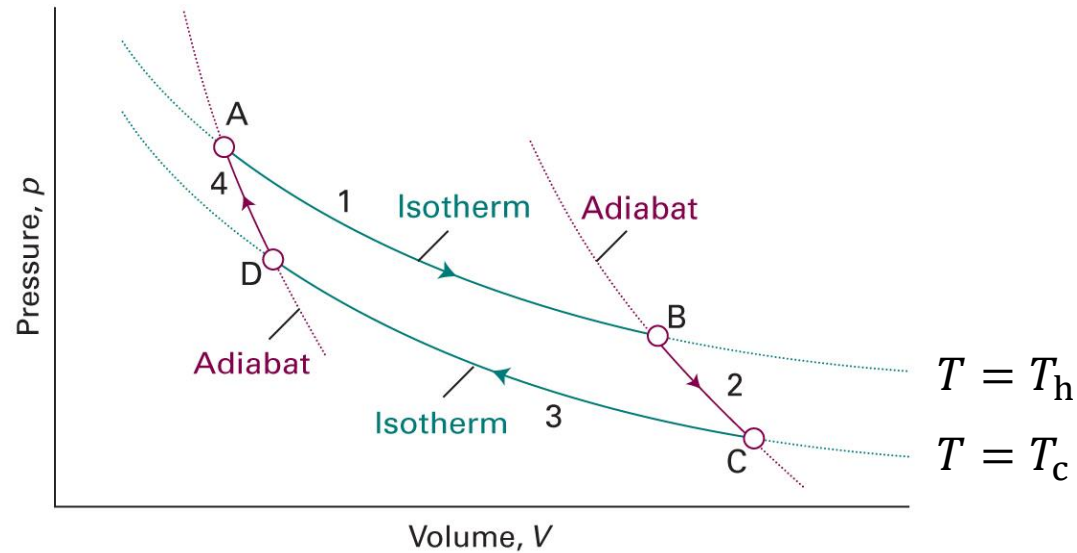
$$q = nRT_h \ln \left(\frac{V_B}{V_A} \right) + nRT_c \ln \left(\frac{V_D}{V_C} \right) = nR(T_h - T_c) \ln \left(\frac{V_B}{V_A} \right)$$

The work is

$$w = -q = -nR(T_h - T_c) \ln \left(\frac{V_B}{V_A} \right)$$

Not in Atkins

Efficiency of Carnot Engine



$$\eta = \frac{|w|}{|q_h|} = \frac{|w|}{|q_1|}$$



(Image: Atkins et al., *Atkins' Physical Chemistry*, 12th ed., Figure 3A.9)

Not in Atkins

Efficiency of Carnot Engine

$$\eta = \frac{|w|}{|q_h|} = \frac{|w|}{|q_1|}$$

For $w = -nR(T_h - T_c) \ln(V_B/V_A)$ and $q_1 = nRT_h \ln(V_B/V_A)$,

$$\eta = \frac{nR(T_h - T_c) \ln(V_B/V_A)}{nRT_h \ln(V_B/V_A)} = \frac{T_h - T_c}{T_h} = 1 - \frac{T_c}{T_h}$$

Carnot's Theorem

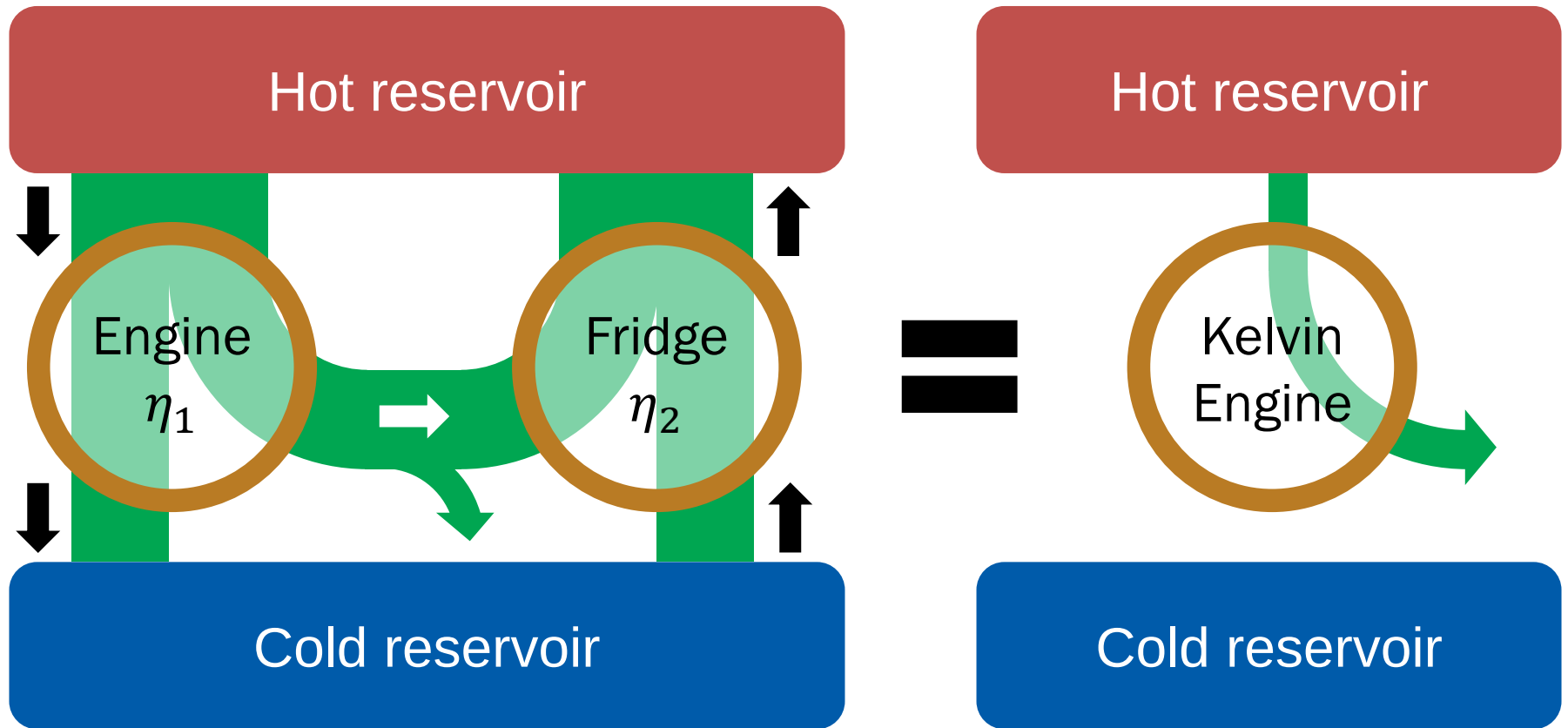
$$\eta = 1 - T_c/T_h \text{ (perfect-gas reversible Carnot engine)}$$

“The efficiency of a reversible Carnot engine depends only on the temperatures of the reservoirs.”

→ independent of the working material

$$\eta = 1 - T_c/T_h \text{ (any reversible Carnot engine)}$$

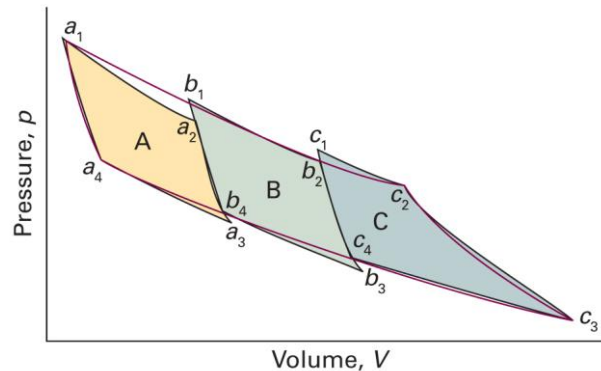
Proof of Equivalence



(if $\eta_1 > \eta_2$)

General Reversible Engine

“All reversible engines have the same efficiency regardless of their construction”



$$\eta = 1 - T_c/T_h \text{ (any reversible engine)}$$

Brief Illustration 3A.3

A certain power station operates with superheated steam at 300 °C ($T_h = 573$ K) and discharges the waste heat into the environment at 20 °C ($T_c = 293$ K). What is its theoretical efficiency?

$$\eta = 1 - \frac{293 \text{ K}}{573 \text{ K}} = 0.489 = 48.9\%$$

One More State Function

$$\eta = 1 - \frac{|q_c|}{|q_h|} = 1 + \frac{q_c}{q_h}$$

For a reversible engine,

$$\eta = 1 - \frac{T_c}{T_h}$$

Hence, for a reversible cycle,

$$\frac{q_h}{T_h} + \frac{q_c}{T_c} = \Delta \left(\frac{q}{T} \right) = 0$$

Entropy

For a reversible cycle,

$$\Delta \left(\frac{q_{\text{rev}}}{T} \right) = 0$$

which implies that q_{rev}/T is a **state function**

Thermodynamic definition of entropy:

$$dS = \frac{\delta q_{\text{rev}}}{T}$$

Entropy Calculation

$$dS = \frac{\delta q_{\text{rev}}}{T}$$

- Entropy change

$$\Delta S = \int_{\text{state } i}^{\text{state } f} \frac{\delta q_{\text{rev}}}{T}$$

- Molar entropy change $\Delta S_{\text{m}} = \Delta S/n$

Example 3A.1

Calculate the entropy change of a sample of perfect gas when it expands isothermally from a volume V_i to a volume V_f .

$$\Delta S = \int_{\text{state } i}^{\text{state } f} \frac{\delta q_{\text{rev}}}{T}$$

Since it is an isothermal change,

$$\Delta S = \frac{1}{T} \int_{\text{state } i}^{\text{state } f} \delta q_{\text{rev}} = \frac{q_{\text{rev}}}{T}$$

For an isothermal change of perfect gas,

$$\Delta U = q + w = 0 \text{ and } w = -nRT \ln(V_f/V_i)$$

Example 3A.1

Since $q = nRT \ln(V_f/V_i)$,

$$\Delta S = \frac{q_{\text{rev}}}{T} = nR \ln\left(\frac{V_f}{V_i}\right)$$

Work for Irreversible Change

For an irreversible change,

$$|\delta w_{\text{irrev}}| < |\delta w_{\text{rev}}|$$

As $\delta w < 0$,

$$\delta w_{\text{irrev}} > \delta w_{\text{rev}}$$

In general,

$$\delta w \geq \delta w_{\text{rev}}$$

Clausius Inequality

$$\delta w \geq \delta w_{\text{rev}}$$

From the first law of thermodynamics,

$$dU = \delta q + \delta w = \delta q_{\text{rev}} + \delta w_{\text{rev}}$$

which leads to

$$\delta q_{\text{rev}} - \delta q = \delta w - \delta w_{\text{rev}} \geq 0$$

Hence,

$$dS = \frac{\delta q_{\text{rev}}}{T} \geq \frac{\delta q}{T}$$

Isolated System

For an isolated system, $\delta q = 0$

$$dS \geq \frac{\delta q}{T} = 0$$

3rd version of the second law of thermodynamics:

“The entropy of an isolated system never decreases.”

Entropy and Spontaneity

For an isolated system,

$$dS \geq 0$$

- Reversible change: $dS = 0$
 - The change can be reversed
 - Achieved only at equilibrium
- Irreversible change: $dS > 0$
 - The change cannot be reversed
 - Arrow of time: spontaneous change

Fundamental Equation

From the first law,

$$dU = \delta q + \delta w$$

For a reversible change,

$$dU = \delta q_{\text{rev}} + \delta w_{\text{rev}} = TdS - pdV$$

As T , p , S , and V are all state functions, in general,

$$dU = TdS - pdV$$

Natural Independent Variables

$$dU = TdS - pdV$$

→ S and V are **natural independent variables** of U

$$U = U(S, V)$$

As U is a state function,

$$dU = \left(\frac{\partial U}{\partial S} \right)_V dS + \left(\frac{\partial U}{\partial V} \right)_S dV$$

Thermodynamic Definitions

$$dU = TdS - pdV$$

As $U = U(S, V)$ is a state function,

$$dU = \left(\frac{\partial U}{\partial S} \right)_V dS + \left(\frac{\partial U}{\partial V} \right)_S dV$$

Comparison leads to

$$T = \left(\frac{\partial U}{\partial S} \right)_V$$

$$p = - \left(\frac{\partial U}{\partial V} \right)_S$$

Examples: Entropy Calculation

Heat transfer

Expansion (reversible vs. free)

Phase transitions

Heating

Heat Transfer

Transfer of heat $|\delta q|$: h $\rightarrow$ c

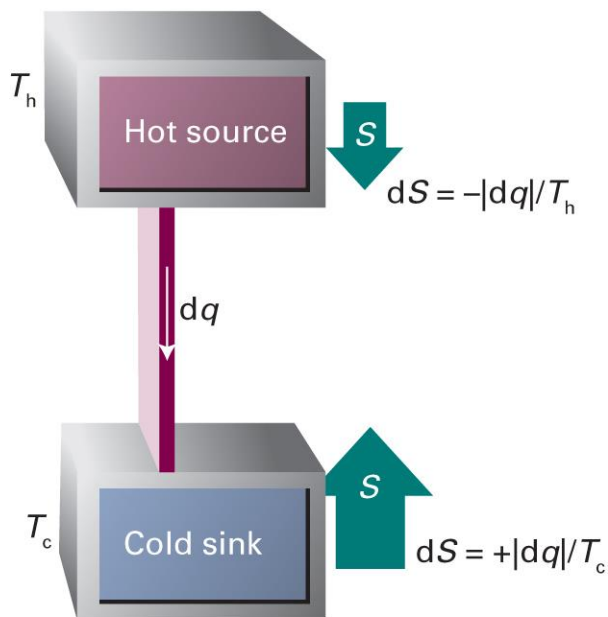
$$dS_h = -|\delta q|/T_h$$

$$dS_c = +|\delta q|/T_c$$

The total entropy change is

$$dS = dS_h + dS_c = |\delta q| \left(\frac{1}{T_c} - \frac{1}{T_h} \right)$$

- Equilibrium: $dS = 0$ or $T_h = T_c$
- Spontaneity: $dS > 0$ or $T_h > T_c$



Heat Transfer to Surroundings

Consider an infinitesimal transfer of heat

$$\delta q = -\delta q_{\text{sur}}$$

The surroundings are extremely large, so

$$\delta q_{\text{sur}} = dU_{\text{sur}} \text{ (constant volume)}$$

$$\delta q_{\text{sur}} = dH_{\text{sur}} \text{ (constant pressure)}$$

Since $dH = dU + d(pV)$, this implies $d(pV)_{\text{sur}} \approx 0$

Entropy of Surroundings

U and H are state functions, so δq_{sur} is independent of reversibility:

$$dS_{\text{sur}} = \frac{\delta q_{\text{sur,rev}}}{T_{\text{sur}}} = \frac{\delta q_{\text{sur}}}{T_{\text{sur}}}$$

As the temperature is invariant,

$$\Delta S_{\text{sur}} = \frac{q_{\text{sur}}}{T_{\text{sur}}}$$

Entropy Calculation

- Heat transfer to surroundings can be always considered as reversible heat transfer

$$\delta q_{\text{sur}} = \delta q_{\text{sur,rev}} \text{ (always)}$$

- Reversible change:

$$dS = \delta q_{\text{rev}}/T = \delta q/T$$

$$dS_{\text{sur}} = \delta q_{\text{sur,rev}}/T = -\delta q/T$$

- Irreversible change:

$$dS \neq \delta q/T \text{ and } dS_{\text{sur}} = -\delta q/T$$

Reversible Expansion

$$\delta q = -\delta q_{\text{sur}}$$

$$\Delta S = \frac{q_{\text{rev}}}{T}$$

$$\Delta S_{\text{sur}} = \frac{q_{\text{sur}}}{T} = -\frac{q_{\text{rev}}}{T} = -\Delta S$$

Hence,

$$\Delta S_{\text{total}} = \Delta S + \Delta S_{\text{sur}} = 0$$

Free Expansion

For a perfect gas, isothermal expansion accompanies with

$$\Delta S = nR \ln \left(\frac{V_f}{V_i} \right)$$

However, $w = 0$ (free expansion) and $\Delta U = 0$ (perfect gas)

$$q = q_{\text{sur}} = 0$$

$$\Delta S_{\text{sur}} = \frac{q_{\text{sur}}}{T} = 0$$

Hence,

$$\Delta S_{\text{total}} = \Delta S + \Delta S_{\text{sur}} > 0$$

Reversible Phase Transition

Normal transition temperature T_{trs} : the temperature at which two phases are in equilibrium at 1 atm

If the system is at T_{trs} and 1 atm,

$$q = \Delta_{\text{trs}}H \text{ (constant pressure)}$$

$$\Delta_{\text{trs}}S = \Delta_{\text{trs}}H/T_{\text{trs}}$$

$$\Delta_{\text{trs}}S_{\text{sur}} = -\Delta_{\text{trs}}H/T_{\text{trs}}$$

$$\Delta_{\text{trs}}S_{\text{total}} = \Delta_{\text{trs}}S + \Delta_{\text{trs}}S_{\text{sur}} = 0$$

Heating at Constant V

$$\Delta S = \int_{\text{state } i}^{\text{state } f} \frac{\delta q_{\text{rev}}}{T}$$

At constant volume, $\delta q_{\text{rev}} = dU = C_V dT$

$$S(T_f) - S(T_i) = \int_{T_i}^{T_f} \frac{C_V}{T} dT$$

For constant C_V ,

$$S(T_f) = S(T_i) + C_V \int_{T_i}^{T_f} \frac{1}{T} dT = S(T_i) + C_V \ln \left(\frac{T_f}{T_i} \right)$$

Heating at Constant p

$$\Delta S = \int_{\text{state } i}^{\text{state } f} \frac{\delta q_{\text{rev}}}{T}$$

At constant pressure, $\delta q_{\text{rev}} = dH = C_p dT$

$$S(T_f) - S(T_i) = \int_{T_i}^{T_f} \frac{C_p}{T} dT$$

For constant C_p ,

$$S(T_f) = S(T_i) + C_p \int_{T_i}^{T_f} \frac{1}{T} dT = S(T_i) + C_p \ln \left(\frac{T_f}{T_i} \right)$$